File permissions in Linux

Project description

This project focuses on examining and modifying file and directory permissions in a Linux environment to ensure that users have only the access they are authorized for. Proper permission management is essential for maintaining system security and preventing unauthorized access.
The tasks were completed using Linux command-line tools and a managed authorization lab environment. The goal was to analyze existing permissions, adjust them according to organizational policies, and document the results.

Check file and directory details

First, I inspected the current permission of files and directories using the **ls -l** command.
This command provides detailed information, including the permission string, the owner and group, file size, modification date and the file or directory name. This information helps identify which items may require permission adjustments.

```
researcher2@153c84045125:~$ cd /home/researcher2/projects
researcher2@153c84045125:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Mar  8 11:23 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Mar  8 11:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar  8 11:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_t.txt
researcher2@153c84045125:~/projects$
```

Describe the permissions string

Breakdown:

- 1st character: **d** for directory and **–** for file type
- Next 3 characters: permission for the user
- Next 3 characters: permission for the group
- Next 3 characters: permission for others

Permission meaning:
r = read
w = write
x = execute
- = no permission

For example, the string **-rw-rw-r--** means:

- User: can read and write
- Group: can read and write
- Other: can only read

Understanding this structure is essential before modifying permissions.

Change file permissions

The organization's policy states that the owner type "others" should not have write access to any files. One file **project_k.txt** needed its permissions adjusted to comply with this rule.

To set the permissions so that the user, group, and others can only read the file, I used:
chmod u=r,g=r,o=r project_k.txt
This command: **chmod o-w project_k.txt** will also execute the same result.

```
researcher2@153c84045125:~$ cd /home/researcher2/projects
researcher2@153c84045125:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Mar  8 11:23 drafts
-rw-rw-rw- 1 researcher2 research_team   46 Mar  8 11:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar  8 11:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_t.txt
researcher2@153c84045125:~/projects$ chmod u=rw,g=rw,o=r project_k.txt
researcher2@153c84045125:~/projects$ ls -l
total 20
drwx--x--- 2 researcher2 research_team 4096 Mar  8 11:23 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar  8 11:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_t.txt
researcher2@153c84045125:~/projects$
```

The result after the command executed is **-rw-rw-r--**
This ensures that the owner type of "other" cannot modify the file, which aligns with the organization's security requirements.

Change file permissions on a hidden file

The research team archived a hidden file named **.project_x.txt** Hidden files in Linux begin with a dot (). The file should not have "write" permissions for anyone, but both the user and the group should be able to read it the file.
To apply the correct permissions:

The command **ls -la** used here to displays permission to files and directories including hidden files.

To apply the correct permission this command is executed:
chmod u=r,g=r .project_x.txt or **chmod u-w,g-w,g+r .project_x.txt**

```
researcher2@153c84045125:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  8 11:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  8 11:36 ..
-rw--w---- 1 researcher2 research_team   46 Mar  8 11:23 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar  8 11:23 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar  8 11:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_t.txt
researcher2@153c84045125:~/projects$ chmod u=r,g=r .project_x.txt
researcher2@153c84045125:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  8 11:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  8 11:36 ..
-r--r----- 1 researcher2 research_team   46 Mar  8 11:23 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar  8 11:23 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar  8 11:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_t.txt
researcher2@153c84045125:~/projects$
```

Now the permission on **.project_x.txt** are: **-r--r-----**
This configuration ensure that owner type “user” and “group” can only read the file.

Change directory permissions

As per policy, the files and directories in the projects directory belong to the **researcher2** user, so only **researcher2** should be allowed to access the **drafts** directory and its contents.
The current string-permission shown that the owner type “group” can execute the file. This means the directory’s permission need to adjusted accordingly.

The command applied here is: **chmod g-x drafts.**
After this change, only the user (researcher2) has full access to the directory.

```
researcher2@153c84045125:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  8 11:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  8 11:36 ..
-r--r----- 1 researcher2 research_team   46 Mar  8 11:23 .project_x.txt
drwx--x--- 2 researcher2 research_team 4096 Mar  8 11:23 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar  8 11:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_t.txt
researcher2@153c84045125:~/projects$ chmod g-x drafts
researcher2@153c84045125:~/projects$ ls -la
total 32
drwxr-xr-x 3 researcher2 research_team 4096 Mar  8 11:23 .
drwxr-xr-x 3 researcher2 research_team 4096 Mar  8 11:36 ..
-r--r----- 1 researcher2 research_team   46 Mar  8 11:23 .project_x.txt
drwx------ 2 researcher2 research_team 4096 Mar  8 11:23 drafts
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_k.txt
-rw-r----- 1 researcher2 research_team   46 Mar  8 11:23 project_m.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_r.txt
-rw-rw-r-- 1 researcher2 research_team   46 Mar  8 11:23 project_t.txt
researcher2@153c84045125:~/projects$
```

Summary

This project demonstrated how to analyze and modify the file and directory permission using linux with the goal to align with organization security policies. The commands used here are:

- **cd:** change directory
- **ls:** display list of files in the current directory
- **ls -l:** Display permission to files & directories
- **ls -la:** Displays permission to files and directories including hidden files
- **chmod:** Changes permissions on files and directories